Retail Sales Forecasting for Multi-Store Retailer

Executive Summary

This project develops a multi-store retail demand forecasting system designed to improve inventory planning and reduce forecast error across stores. Using machine learning techniques and time-series feature engineering, the final XGBoost model achieved a 57% reduction in RMSE compared to a naive baseline and reached a MAPE of 9.75%.

Business Objective

The goal of this project is to accurately forecast daily sales per store to support inventory optimization, reduce stockouts and overstock, and improve operational efficiency in a retail environment.

Dataset Description

- Date: Daily timestamp of sales
- Store Number: Unique store identifier
- Product Family: Product category
- Sales: Target variable (daily sales)
- On Promotion: Number of items under promotion

Methodology

- Aggregated sales at store-day level
- Created time-based features (month, day of week, week of year)
- Engineered lag features (lag_1, lag_7, lag_30)
- Created rolling averages (7-day and 30-day windows)
- Time-based train/test split (last 3 months as test set)

Models Implemented

- Naive Baseline (Lag-1 Forecast)

- Random Forest Regressor
- XGBoost Regressor

Model Performance

- Baseline RMSE: 5348
- Random Forest RMSE: 2485
- XGBoost RMSE: 2278
- MAPE: 9.75%
- RMSE Reduction vs Baseline: 57%

Business Impact

By reducing forecast error by 57%, the model significantly improves demand planning accuracy. Lower forecasting error directly contributes to reduced safety stock requirements, fewer stockouts, and improved supply chain efficiency. Achieving sub-10% MAPE demonstrates production-ready retail forecasting capability.

Conclusion

This project demonstrates the application of machine learning for retail demand forecasting. Through feature engineering and advanced modeling techniques, the system delivers strong predictive performance and provides measurable business value for inventory optimization and operational planning.